FHIR-STU3 — FHIR Linked Data Module (sections 3 – 3.3)

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3 FHIR Linked Data Module

URL: https://hl7.org/fhir/STU3/linked-data-module.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Work Group Implementable Technology Specifications | Ballot Status: Informative

3.0 FHIR Linked Data Module

3.0.2 Introduction

This module describes the RDF representation for FHIR resources (FHIR/RDF) and related assets, including an OWL ontology for FHIR/RDF and a ShEx grammar to validate FHIR/RDF. Linked Data is structured data that is represented in an RDF format to facilitate inference and data linkage across datasets.

Materials in this module are created and maintained by a collaboration between HL7 and W3C . Editor: David Booth

3.0.1.1 Motivation and Design of the FHIR Linked Data Module

Although RDF formats can be used to exchange FHIR data, the primary purpose of the FHIR Linked Data Module is to ground FHIR data semantics in RDF, as a universal information representation. This enables FHIR data to be used with RDF applications, to support inference, shared semantics across multiple standards and data formats, data integration, semantic data validation, compliance enforcement, SPARQL queries and other uses.

The FHIR/RDF is based on the same abstract FHIR information model as the XML and JSON formats, and carries the same FHIR information content. FHIR data is losslessly round-trippable between XML, JSON and RDF formats. For this reason, data expressed in FHIR/RDF corresponds closely to the XML and JSON formats in its look and feel. However, there are a few additional terms that may appear in the RDF format, which are explained on the RDF page: fhir:nodeRole, fhir:treeRoot, fhir:value, fhir:index and fhir:concept.

Ontologies that were designed independently almost always have some impedance mismatch when attempting to use them together. Many of the ontologies in the medical and life sciences domain are designed to capture facts about the world for research, such as the fact that the mitral valve is a kind of heart valve. But FHIR was designed to support the day-to-day operations of healthcare providers exchanging electronic health records (EHRs), and in this context the orientation has historically been different. When using FHIR/RDF with other ontologies, impedence differences are likely to show up in two main ways:

* Records versus facts. FHIR is oriented toward recording who did what ("Dr. Jones diagnosed patient x with viral pneumonia") rather than stating absolute medical facts ("patient x has viral pneumonia").

* Non-monotonicity. RDF was designed to be monotonic, whereas FHIR has a few design aspects that are would be non-monotonic if they were interpreted directly in RDF. (Monotonicity means that new data cannot invalidate previous conclusions; non-monotonicity means that previous conclusions can be invalidated by new data.) For example, a modifier extension indicates that the surrounding element's meaning will likely be misunderstood if the modifier extension is not understood.

For both of these reasons, to maintain monotonicity in RDF, FHIR/RDF should not be directly interpreted as stating facts, at least until any potentially non-monotonic elements have been removed or isolated through pre-processing.

Application developers should also be aware that some FHIR data attributes have a major impact on the interpretation of the enclosing data element: the meaning of the enclosing element cannot be determined in isolation. For example, a status of 'entered-in-error' means that the resource was created accidentally, and should be ignored.

3.0.1.2 FHIR/RDF Data Formats

FHIR/RDF examples are provided for all FHIR resources in both Turtle and JSON-LD formats. To access a JSON-LD example, first navigate to that example in Turtle format, and then click the "JSON-LD" link above the Turtle example.

The mime type for the Turtle format is text/turtle . The mime type for the JSON-LD format is application/ld+json .

3.0.2 Using FHIR/RDF

Application developers wishing to use FHIR/RDF will often need to perform the following rough steps, though exact steps will depend on your application:

* Convert FHIR/XML or FHIR/JSON data to/from FHIR/RDF. Open source implementations that can perform this conversion include:

* HAPI java Reference library (Same code as used in the build)

* Reference Server (based on Delphi reference implementation)

* fhir-net-api , a .NET implementation. (should work on Mono)

* Check the HL7 Open Source FHIR Implementations page, in case new implementations have become available.

* Download the FHIR ontology.

* Download other ontologies that you wish to use with your application, such as the SNOMED-CT ontology. (Because of license restrictions, the process of obtaining the SNOMED-CT ontology is more complex than a simple download. Some guidance is provided here .)

* Create or obtain a bridge ontology that relates terms in the FHIR ontology to terms in your other ontologies.

* Load your FHIR/RDF instance data and your ontologies into a reasoner and/or a triplestore, and perform SPARQL queries and/or reasoning.

* If you are creating FHIR/RDF then you might also wish to validate it, using ShEx . If so, you can download the ShEx schema for FHIR/RDF.

Slides by Eric Prud'hommeaux illustrate how inference can be used on FHIR/RDF data to perform a query for FHIR Observations of rheumatoid arthritis, using the SNOMED-CT ontology . Caveat: those slides show some terms in the fhir: namespace that do not actually exist in the fhir: namespace, such as fhir:Observation-of-Rheumatoid_arthritis-disorder and fhir:Coding-of-Rheumatoid_arthritis-disorder. A different namespace should have been used in the example.

3.0.3 Index

Other content in this module:

* FHIR/RDF Introduction See also: Additional FHIR/RDF Introductory materials, not yet edited or merged

* (Draft) FHIR/RDF formal specification (at W3C)

* Downloadable ShEx Validation Schema for FHIR RDF

* Downloadable FHIR Ontology

3.0.4 Security and Privacy

The use of RDF does not raise any known security or privacy issues beyond those already existing for FHIR/XML or FHIR/JSON. For more general considerations, see the Security and Privacy module.

RDF can also be used to help enforce security and privacy policies. See also HL7 Security and Privacy Ontology.

3.0.5 Common use Cases

In this section: common problems in the space of the module, ways to go about solving them, or references to additional problem based links.

TODO

3.0.6 Developmental Roadmap

In this section: what the current overall state is, what work is in train, what the goals over the next 18 months or so are.

Most of the major pieces of FHIR/RDF are now available in a complete enough form for developers to start trying them and providing feedback. Work still needed:

* Further editing of FHIR Linked Data module pages.

* FHIR Ontology Introductory Page. This still needs to be drafted, perhaps based the FHIR Ontology draft on W3C github .

* Tutorial on using FHIR/RDF with other ontologies (such as the SNOMED-CT ontology) and performing inference. Eric Prud'hommeaux's slides or Tony Mallia's AllergyIntolerance example may be a good starting point.

* Tutorial on using ShExMap to translate other RDF data (non-FHIR) to/from FHIR/RDF.

3.1 Resource Description Framework (RDF) Representation

URL: https://hl7.org/fhir/STU3/rdf.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

3.1 Resource Description Framework (RDF) Representation

FHIR Infrastructure Work Group | Maturity Level: 2 | Ballot Status: Trial Use

| This page and the RDF forms are jointly maintained by the HL7 FHIR project and the W3C Semantic Web Health Care and Life Sciences Interest Group .

FHIR resources can be represented as an RDF graph serialised in the Turtle format , or as JSON-LD . The RDF format is defined to assist the process of bridging between operational data exchange and formal knowledge processing systems. While the RDF form offers a fully functional representation of FHIR resources, it has different operational characteristics to the JSON and XML representations, and would be implemented for different reasons. Systems focused on operational exchange of data would not generally use choose to use RDF.

This page describes:

* Representing FHIR resources in Turtle

* The FHIR definitions as Turtle

* How to exchange RDF using the RESTful API

* Experimental JSON-LD format

3.1.1 Turtle Format for Resources

A FHIR resource is represented by a series of RDF triples. The Turtle representation for a resource is described using this format:

```
Turtle Template

[ a fhir:Observation; fhir:nodeRole fhir:treeRoot;
  # from Resource: id; meta; implicitRules; and language
  # from DomainResource: text; contained; extension; and modifierExtension
  fhir:Obervation.identifier [ Identifier ]; # 0..* Unique Id for this particular observation
  fhir:Obervation.status [ fhir:value "<code>" ]; # R!  registered | preliminary | final | amended +
  fhir:Obervation.code [ CodeableConcept ]; # 1..1 R!  Type of observation (code / type)
  fhir:Obervation.subject [ fhir:reference [ Patient|Group|Device|Location ] ]; # 0..1 Who and/or what this is about
  fhir:Obervation.encounter [ fhir:reference [ Encounter ] ]; # 0..1 Healthcare event during which this observation is
    made
  # effective[x]: 0..1 Clinically relevant time/time-period for observation. One of these 2:
    fhir:Obervation.effectiveDateTime [ fhir:value "<dateTime>" ];
    fhir:Obervation.effectivePeriod [ Period ];
]
```

Using this format:

* To build a valid Turtle instance of a resource, replace the contents of the property values with valid content as described by the type rules and content description found in the property value for each element

* Relationship names are case-sensitive (though duplicates that differ only in case are never defined)

* Relationships can appear in any order

* Content within a resource is always represented with anonymous nodes - only resources can be identified

* Nodes are never empty. If an element is present in the resource, it SHALL have properties as defined for its type, or 1 or more extensions

* The R! denotes that a relationship is mandatory - it must be present (or in an array, at least one item must be present)

* Note that this specification produces turtle that is nicely formatted and well laid out, but this is not required nor expected

* The MIME-type for this format is text/turtle. Other MIME types: application/json-ld for the json-ld variant, and text/shex for the SHEX (RDF schemas)

3.1.1.1 Representing resources

Each resource is represented as a set of RDF triples represented using the Turtle syntax. When a resource has a persistent identity (e.g. it can be found at a particular URL - usually a FHIR RESTful server), then that URL is its identity. Resources with no persistent identity (e.g. bundles from search results) have the the identity of the root document - "<>" in turtle syntax.

Some resources can contain other resources. Given that the relationships can appear in any order, it cannot be assumed that the first encountered element represents the resource of interest that is being represented by the set of turtle statements. The focal resource - where to start when parsing - is the resource with the relationship fhir:nodeRole to fhir:treeRoot. if there is more than one node labeled as a 'treeRoot' in a set of turtle statements, it cannot be determined how to parse them as a single resource.

3.1.1.2 Representing Repeating Elements

Elements that can repeat are represented with a relationship

```
  fhir:index [n]
```

where [n] is a zero-based integer offset (ie.g the first element has an index of 0). Lists are never sparse; it is an error if there are missing items in the repeat sequence.

Note: this measn that the rdf:list structure is not used.

3.1.1.3 Representing Primitive Elements

Primitive elements - elements with a primitive type - are represented as regular nodes so that the elements extensions can be represented. The actual value of the primitive type is represented using the fhir:value predicate:

```
  fhir:value "[value]"^^xs:type
```

The value has two parts: a literal string that contains the value, and, if applicable, one of the following schema types:

* boolean

* integer

* decimal

* base64Binary

* dateTime

* date

* gYear

* gYearMonth

* time

The choice is made based on the types as specified for the primitive type. Note that the correct schema type for a date/dateTime must be determined by inspecting the value of the date for precision.

The fhir:value property can never be empty. Either the relationship is absent, or it is present with at least one character of content. XHTML is represented as an escaped xs:string.

3.1.1.4 Representing References

A Reference element is representing using the same rules as above:

```
fhir:Observation.subjectReference [
    fhir:Reference.reference [ fhir:value "Patient/example" ];
    fhir:Reference.display [ fhir:value "Example Patient" ];
  ];
```

This allows faithful round tripping of the resource between the Turtle format and the JSON and XML formats. However, it's very useful for an RDF processor if the RDF graph links to the target of the reference directly. This can be represented using the fhir:link property:

```
fhir:Observation.subjectReference [
    fhir:link <http://hl7.org/fhir/Patient/example>;
    fhir:Reference.reference [ fhir:value "Patient/example" ]
  ];
```

The correct value for the fhir:link relationship must be determined by resolving the rules for resolving references for the various reference types to a literal URL that refers to the correct content in the local RDF context.

The fhir:link relationship can be added automatically as part of generating the resource representation, or it can be injected by a post-processor that knows how to convert the raw references into RDF-suitable references.

3.1.1.5 Representing Concepts

The same logic applies to the Coding data type. These are represented directly in turtle by serialising their properties as specified above:

```
  fhir:Observation.code [
     fhir:CodeableConcept.coding [
       fhir:index 0;
       fhir:Coding.system [ fhir:value "http://loinc.org" ];
       fhir:Coding.code [ fhir:value "29463-7" ];
       fhir:Coding.display [ fhir:value "Body Weight" ]
    ];
    fhir:CodeableConcept.coding [
       fhir:index 1;
       fhir:Coding.system [ fhir:value "http://snomed.info/sct" ];
       fhir:Coding.code [ fhir:value "27113001" ];
       fhir:Coding.display [ fhir:value "Body weight" ]
    ]
  ];
```

For reasoners using the RDF graph, it's very useful to make the implicit concept references in these Codings explicit using a rdf:type assertion ("a" in turtle):

```
  fhir:Observation.code [
     fhir:CodeableConcept.coding [
       fhir:index 0;
       a loinc:29463-7;
       fhir:Coding.system [ fhir:value "http://loinc.org" ];
       fhir:Coding.code [ fhir:value "29463-7" ];
       fhir:Coding.display [ fhir:value "Body Weight" ]
     ];
     fhir:CodeableConcept.coding [
       fhir:index 2;
       a sct:27113001;
       fhir:Coding.system [ fhir:value "http://snomed.info/sct" ];
       fhir:Coding.code [ fhir:value "27113001" ];
       fhir:Coding.display [ fhir:value "Body weight" ]
    ]
  ];
```

These rdf:type assertions can be made by any agent that knows how to convert from the code system to the correct ontological representation on the RDF context. Note that a few code systems have standard ontological representations, but many don't. Again, these assertions can be made by the serialiser, or injected by a post-processor.

3.1.1.6 Schema

FHIR uses ShEx for representing the turtle schema. See fhir.shex for definitions.

3.1.1.7 Experimental JSON-LD format

As a by-product of the RDF format, a JSON-LD format is defined to represent exactly the same information as the Turtle format, but using the JSON-LD syntax. Some specific notes about the JSON-LD format:

* There is a single FHIR context: [url]/fhir.jsonld, where [url] is the URL of the applicable version (e.g. http://hl7.org/fhir/STU3). Implementers should use a version specific reference, not the current version (http://hl7.org/fhir) since the contents of this change as new full releases of FHIR are published)

* Each resource has @type on every resource, since the resources are inherently polymorphic at the format level. In addition, the root node explicitly carries the property "role" : "fhir:treeRoot"

* When parsing the JSON-LD format, applications should not use the literal JSON directly, but use the json-ld processed form. Any json-ld that results in the same content post-processing is considered conformant.

* The JSON-LD format is derived from the RDF format, and processing the JSON-LD will result in the same set of triples as the turtle format, though there may be minor differences in the types of the nodes depending on the RDF library used.

* For booleans and integers, JSON boolean and numbers are used. However for decimal values, a JSON string must be used.

STU Note: The definition of the JSON-LD format means that the FHIR specification defines two different JSON formats, each with a different purpose. It seems unlikely that normal implementers will prefer the JSON-LD format over the other JSON format, since that is more dense, but there are other implementers who would prefer the JSON-LD format. On the other hand, defining two different JSON formats seems like a recipe for confusion. Feedback from implementers is welcome.

Feedback is welcome here .

3.1.2 RDF Representation of FHIR

In addition to the basic representation of FHIR resources in turtle format, a Turtle representation of the FHIR infrastructure and definitions is also published, for the following purposes:

* Providing the class definitions to support RDF based representation of resource instances

* Supporting knowledge based analysis of the FHIR specification itself

* Providing knowledge of use at run-time for converting between FHIR and other content models

* Supporting reasoning across the information/terminology model boundary

The RDF definitions are published as a series of turtle files: RIM and FHIR. Note: these are out of sync with the serialization above; this will be fixed in the lead up to the Montreal connectathon.

3.1.3 Using RDF with the REST API

TODO

3.2 FHIR Status Codes Grid

URL: https://hl7.org/fhir/STU3/sc.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

This is an internal QA page. The intent of this table is to help reviewers compare the usage of status codes across resources: where the status code means the same thing, it should have the same code. The columns have no formal definition; their intent is to be inferred from the codes listed in the column.

Note: This page is not proscriptive; it's descriptive and comparative, to assist with review. The FHIR methodology says that if two domains use different terms for the same concept (e.g. 'withdrawn' and 'cancelled') then the domain concept name should be used. Conversely, where there is no explicit reason for the codes to differ based on domain usage, they should not.

Note: Bold indicates that the element is marked as a modifier.

Path | c1 | c2 | c3 | c4 | c5 | c6 | c7 | c8 | c9 | c10 | c11 | c12 | c13 | c14 | c15 | c16 | c17 | c18 | c19 | c20 | c21 | c22 | c23 | c24 | c25 | c26 | c27 | c28 | c29 | c30 | c31 | c32 | c33 | c34 | c35 | c36 | c37 | c38 | c39 | c40 | c41 | c42 | c43 | c44 | c45 | c46 | c47 | c48 | c49 | c50 | c51 | c52 | c53 | c54 | c55 | c56 | c57 | c58 | c59 | c60 | c61 | c62 | c63 | c64 | c65 | c66 | c67 | c68 | c69 | c70 | c71 | c72 | c73 | c74 | c75 | c76 | c77 | c78 | c79 | c80

Account.status | entered-in-error | | | | | | | | active | | | | | inactive

ActivityDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown

AllergyIntolerance.clinicalStatus | | | | | | | | | | active | | | | | inactive | | | | | resolved

AllergyIntolerance.verificationStatus | entered-in-error | | | | | | | | | | | | | | | | | unconfirmed | confirmed | | refuted

Appointment.participant.status | entered-in-error | | | tentative | | | declined | accepted | | | | needs-action

Appointment.status | entered-in-error | proposed | | pending | | | | booked | arrived | | noshow | | | fulfilled | | cancelled

AppointmentResponse.participantStatus | entered-in-error | | | tentative | | | declined | accepted | | in-process | | | | completed | | | needs-action

CapabilityStatement.status | | | | draft | | | | | | active | | | | | retired | | unknown

CarePlan.activity.detail.status | | | | not-started | | | | scheduled | | in-progress | on-hold | | | completed | | cancelled | unknown

CarePlan.status | entered-in-error | proposed | | draft | | | | | | active | suspended | | | completed | | cancelled | unknown

CareTeam.status | entered-in-error | proposed | | | | | | | | active | suspended | | | | inactive

ChargeItem.status | entered-in-error | | planned | | | | not-billable | | | billable | | aborted | | billed | | | unknown

Claim.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled

ClaimResponse.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled

ClinicalImpression.status | entered-in-error | | | | | | | | | draft | | | | completed

CodeSystem.status | | | | draft | | | | | | active | | | | | retired | | unknown

Communication.status | entered-in-error | | preparation | | | | | | | in-progress | suspended | aborted | | completed | | | unknown

CommunicationRequest.status | entered-in-error | | | draft | | | | | | active | suspended | | | completed | | cancelled | unknown

CompartmentDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown

Composition.status | entered-in-error | | | preliminary | | | | | | | | amended,appended | final

ConceptMap.status | | | | draft | | | | | | active | | | | | retired | | unknown

Condition.verificationStatus | entered-in-error | | | | | | | | | | | | | | | | | unknown | working, provisional | confirmed | | refuted | differential

Consent.status | entered-in-error | proposed | | draft | | | rejected | | | active | | | | | inactive

Coverage.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled

DataElement.status | | | | draft | | | | | | active | | | | | retired | | unknown

DetectedIssue.status | entered-in-error | | | preliminary | | registered | | | | | | | corrected, amended | final | | cancelled | unknown

Device.status | entered-in-error | | | | | | | | active | | | | | inactive | | unknown
DeviceComponent.operationalStatus | entered-in-error | | | | | | | | | on | standby | | | | off | | | | | | | | | | | | | | not-ready | transduc-discon | hw-discon
DeviceMetric.operationalStatus | entered-in-error | | | | | | | | on | standby | | | | off
DeviceRequest.status | entered-in-error | | | draft | | | | | | active | suspended | | | completed | | cancelled
DeviceUseStatement.status | entered-in-error | | intended | | | | | | | active | on-hold | stopped | | | completed
DiagnosticReport.status | entered-in-error | | | partial,preliminary | | | registered | | | | | | | corrected, amended, appended | final | | cancelled | unknown
DocumentManifest.status | entered-in-error | | | | | | | | | current | | | | | superseded
DocumentReference.docStatus
DocumentReference.status | entered-in-error | | | | | | | | | current | | | | | superseded
EligibilityRequest.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
EligibilityResponse.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
Encounter.location.status | | | planned | | | | | reserved | | active | | | | completed
Encounter.status | entered-in-error | unknown | planned | | | | | triaged | arrived | in-progress | onleave | | | finished | | cancelled
Encounter.statusHistory.status | entered-in-error | unknown | planned | | | | | triaged | arrived | in-progress | onleave | | | finished | | cancelled
Endpoint.status | entered-in-error | | | test | | | | | | active | suspended | error | | off
EnrollmentRequest.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
EnrollmentResponse.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
EpisodeOfCare.status | entered-in-error | | planned | waitlist | | | | | | active | onhold | | | finished | | cancelled
EpisodeOfCare.statusHistory.status | entered-in-error | | planned | waitlist | | | | | | active | onhold | | | finished | | cancelled
ExpansionProfile.status | | | | draft | | | | | | active | | | | | retired | | unknown
ExplanationOfBenefit.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
FamilyMemberHistory.status | entered-in-error | | | | | | | | | | | | | completed | | | health-unknown | | | | | | partial
Flag.status | entered-in-error | | | | | | | | | active | | | | | inactive
Goal.status | entered-in-error | proposed | planned | | | | rejected | accepted | | in-progress, sustaining | on-hold | | | achieved | | cancelled | | | | | | | | | | on-target | ahead-of-target | behind-target
GraphDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown
GuidanceResponse.status | entered-in-error | | | data-requested | | | data-required | | | in-progress | | failure | | success
Immunization.status | entered-in-error | | | | | | | | | in-progress | on-hold | stopped | | completed
ImplementationGuide.status | | | | draft | | | | | | active | | | | | retired | | unknown
Library.status | | | | draft | | | | | | active | | | | | retired | | unknown
List.status | entered-in-error | | | | | | | | | current | | | | | retired
Location.status | | | | | | | | | | active | suspended | | | | inactive
Measure.status | | | | draft | | | | | | active | | | | | retired | | unknown
MeasureReport.status | | | | | | | | | | pending | | | error | | complete
Medication.status | entered-in-error | | | | | | | | | active | | | | | inactive
MedicationAdministration.eventHistory.status | entered-in-error | | | | | | | | | in-progress | on-hold | stopped | | | completed
MedicationAdministration.status | entered-in-error | | | | | | | | | in-progress | on-hold | stopped | | completed | | | unknown
MedicationDispense.eventHistory.status | entered-in-error | | | | | | | | | in-progress | on-hold | stopped | | completed
MedicationDispense.status | entered-in-error | | preparation | | | | | | | in-progress | on-hold | stopped | | completed
MedicationRequest.eventHistory.status | entered-in-error | | | draft | | | | | | active | on-hold | stopped | | completed
MedicationRequest.status | entered-in-error | | | draft | | | | | | active | on-hold | stopped | | completed | | cancelled | unknown
MedicationStatement.status | entered-in-error | | intended | | | | | | | active | on-hold | stopped | | completed
MessageDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown
NamingSystem.status | | | | draft | | | | | | active | | | | | retired | | unknown
NutritionOrder.status | entered-in-error | proposed | planned | draft | requested | | | | | active | on-hold | | | completed | | cancelled
Observation.status | entered-in-error | | | preliminary | | registered | | | | | | | corrected, amended | final | | cancelled | unknown
OperationDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown
PaymentNotice.status | entered-in-error | | | draft | | | | | | active | | | | | cancelled
PaymentReconciliation.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
PlanDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown
Procedure.status | entered-in-error | | preparation | | | | | | | in-progress | suspended | aborted | | completed | | | unknown
ProcessRequest.status | entered-in-error | | | draft | | | | | | active | suspended | | | completed | | cancelled | unknown
ProcessResponse.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled
Questionnaire.status | | | | draft | | | | | | active | | | | | retired | | unknown
QuestionnaireResponse.status | entered-in-error | | | | | | | | | in-progress | | stopped | amended | completed
ReferralRequest.status | entered-in-error | | | draft | | | | | | active | | | | completed | | cancelled
RequestGroup.status | entered-in-error | | | draft | | | | | | active | suspended | | | completed | | cancelled | unknown
ResearchStudy.status | entered-in-error | | | draft | | | | | | in-progress | suspended | stopped | | | completed
ResearchSubject.status | entered-in-error | | | candidate | | | | | enrolled | active | suspended | withdrawn | | completed
SearchParameter.status | | | | draft | | | | | | active | | | | | retired | | unknown
ServiceDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown
Slot.status | entered-in-error | | | | | | | busy | | | | | | | | | busy-tentative | | | | | | busy-unavailable | free
Specimen.status | entered-in-error | | | | | | | | | available | | unsatisfactory | | | unavailable
StructureDefinition.status | | | | draft | | | | | | active | | | | | retired | | unknown
StructureMap.status | | | | draft | | | | | | active | | | | | retired | | unknown
Subscription.status | entered-in-error | | | | requested | | | | | active | | error | | | off
Substance.status | entered-in-error | | | | | | | | active | | | | | inactive
SupplyDelivery.status | entered-in-error | | | | | | | | | in-progress | | | | completed | | abandoned
SupplyRequest.status | entered-in-error | | | draft | | | | | | active | suspended | | | completed | | cancelled | unknown
Task.status | entered-in-error | | | draft | requested | received | rejected | accepted | | in-progress | on-hold | failed | | completed | | cancelled | | | | | | | | | ready
TestReport.status | entered-in-error | | | | | | | | in-progress | waiting | stopped | | completed

TestScript.status | | | | draft | | | | | | active | | | | | retired | | unknown
ValueSet.status | | | | draft | | | | | | active | | | | | retired | | unknown
VisionPrescription.status | entered-in-error | | | draft | | | | | | active | | | | | | cancelled

3.3 FHIR W5 Report

URL: https://hl7.org/fhir/STU3/w5.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Concise

Resource | id | id.version | status | class | grade | what | who.focus | context | when.init | when.planned | when.done | when.recorded | who.author | who.source | who.actor | who.cause | who.witness | who | where | why

Clinical Resources

ActivityDefinition (2) | url identifier* | version | status | experimental | | | | | | | date | | | | | publisher | | | purpose

AdverseEvent (0) | | | | category | | type | | | date | | | | recorder | | | | | subject

AllergyIntolerance (3) | identifier* | | clinicalStatus verificationStatus | type category* | criticality | code | patient | | onset[x] | | | | assertedDate | recorder | asserter

BodySite (1) | identifier* | | active | | | code qualifier* description image* | patient

CarePlan (2) | identifier* | | status | category* | | description | subject | context | period | | | | | author* | careTeam* | | | | | addresses*

CareTeam (2) | identifier* | | status | category* | | | subject | context | period | | | | | | member | | | | | reasonCode* reasonReference*

ClinicalImpression (0) | identifier* | | status | | | code | subject | context | | | effective[x] | date | assessor | | | | | | | problem*

Condition (3) | identifier* | | clinicalStatus verificationStatus | category* | severity | code | subject | context | onset[x] | | abatement[x] | assertedDate | asserter | | | | | | | code* detail*

DetectedIssue (1) | identifier | | status | category | severity | | patient | | | | | date | author

DiagnosticReport (3) | identifier* | | status | category | | code | subject | context | | | effective[x] | issued | | | | | performer*

FamilyMemberHistory (2) | identifier* | | status | | | | patient | | | | | date | | | | | | | | reasonCode* reasonReference*

Goal (2) | identifier* | | status | category* | priority | description | subject | | | start[x] | due[x] | statusDate | | expressedBy | | | | | | addresses*

GuidanceResponse (2) | identifier | | status | | | | subject | context | | | occurrenceDateTime | | | | performer | | | | | reason[x]

ImagingManifest (1) | identifier | | | | | | patient | | | | | authoringTime | author

ImagingStudy (3) | uid accession identifier* | | availability | modalityList* | | | patient | context | started | | | | | | performer* | basedOn* referrer | interpreter* | | | reason

Immunization (3) | identifier* | | status notGiven | | | vaccineCode | patient | encounter | date | | | | | primarySource reportOrigin | actor | | | | location

ImmunizationRecommendation (1) | identifier* | | | | | | patient

Library (2) | url identifier* | version | status | experimental | | type | | | | | | date | | | | | publisher | | | purpose

Measure (2) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose

MeasureReport (2) | identifier | | status | type | | measure | patient | | | | | date | | reportingOrganization

Medication (3) | | | | code isBrand isOverTheCounter | | | | | | | | | | | manufacturer

MedicationAdministration (2) | identifier* | | status | | | medication[x] | subject | context supportingInformation* | | | effective[x] | | | | performer* | | | | | reasonReference*

MedicationDispense (2) | identifier* | | status | | | medication[x] | subject | supportingInformation*

MedicationRequest (3) | identifier* | | status | intent category | priority | medication[x] | subject | context supportingInformation* | | | | authoredOn | requester | | | | | recorder performer | | reasonCode* reasonReference*

MedicationStatement (3) | identifier* | | status | category | | medication[x] | | | | | effective[x] | dateAsserted | | informationSource | | | | subject | | reasonCode* reasonReference*

NutritionOrder (2) | identifier* | | status | | | type* modifier type baseFormulaType additiveType | patient | encounter | | | | dateTime | orderer

Observation (5) | identifier* | | status | category* | | code code | subject | context | | | effective[x] | issued | | | performer*

PlanDefinition (2) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose

Procedure (3) | identifier* | | status | category | | code | subject | context | | | performed[x] | | | | actor | | | | location | reasonCode* reasonReference*

ProcedureRequest (3) | identifier* | | status | intent priority category* | | code | subject | context | | occurrence[x] | | authoredOn | agent | | performerType performer | | | | | reasonCode* reasonReference*

ReferralRequest (1) | identifier* | | status | intent type | priority | serviceRequested* | subject | context | | occurrence[x] | | authoredOn | agent | | recipient* | | | | | reasonCode* reasonReference*

RequestGroup (0) | identifier* | | status | intent | priority | | subject | context | | | | authoredOn | author | | participant* | | | | | reason[x]

ResearchStudy (0) | identifier* | | status | | | focus* | enrollment* | | | | period | | | | sponsor principalInvestigator | | | | site* | reasonStopped

ResearchSubject (0) | identifier | | status | | | | | | | period

RiskAssessment (1) | identifier | | status | method | | code | subject | context | | | occurrence[x] | | | | performer | | | | | reason[x]

Sequence (1) | identifier*

ServiceDefinition (0) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose

Specimen (2) | identifier* accessionIdentifier | | status | | | type | subject | | collected[x] | | receivedTime | | | | collector | | | | | request*

VisionPrescription (1) | identifier* | | status | | | | patient | encounter | | | | dateWritten | prescriber | | | | | | | reason[x]

Administrative Resources

Device (2) | identifier* udi | | status | | | deviceIdentifier name type lotNumber manufacturer manufactureDate expirationDate model version | patient | | | | | | owner contact* | | | | | location url

DeviceComponent (1) | identifier | | | type | | | | | lastSystemChange

DeviceMetric (1) | identifier | | | type category

Group (1) | identifier* | | | active | type actual | | | code

HealthcareService (2) | identifier* | | | active | category | | | | | | | | | | | | | | | location*

Location (3) | identifier* | | status operationalStatus | mode type physicalType

Organization (3) | identifier* | | | active | type*

Patient (5) | identifier* | | active

Person (2) | identifier* | | active

Practitioner (3) | identifier* | | active

PractitionerRole (2) | identifier* | | active | | | | | | | period | | | | | | | | location*
RelatedPerson (2) | identifier* | | active | relationship | | | | | | | period
Substance (2) | identifier* | | status | category* | | | code
Workflow Resources
Appointment (3) | identifier* | | status | specialty* priority | | | | supportingInformation* | start | | end | | | | | | | actor
AppointmentResponse (3) | identifier* | | | | | | | start | | end | | | | | | | actor
Communication (2) | identifier* | | status | category* | | | subject recipient* | topic* context | sent | | received | | | | sender | | | | | reasonCode* reasonReference*
CommunicationRequest (2) | identifier* | | status | category* | priority | | subject recipient* | topic* context | | occurrence[x] | | authoredOn | agent | | sender | | | | | reasonCode* reasonReference*
DeviceRequest (0) | identifier* | | status | intent | priority | code[x] | subject | context | | occurrence[x] | | authoredOn | agent | | performerType performer | | | | | reasonCode* reasonReference*
DeviceUseStatement (0) | identifier* | | status | | | | subject | | | | whenUsed timing[x] | recordedOn | | | source device | | | | | indication*
Encounter (2) | identifier* | | status | class type* | priority | | subject | episodeOfCare* | | | period | | | | | | | individual | location | reason* condition
EpisodeOfCare (2) | identifier* | | status | type* | | condition | patient | | period
Flag (1) | identifier* | | status | category | | code | subject | period encounter | | | | | author
ProcessRequest (2) | identifier* | | status | action | | | target | | | | | created | | provider organization
ProcessResponse (2) | identifier* | | status | | | | | | | | created | | organization | | | | | request
Schedule (3) | identifier* | | active | serviceCategory serviceType* specialty* | | | actor* | | | planningHorizon
Slot (3) | identifier* | | | serviceCategory serviceType* specialty* appointmentType | | | | | | start end
SupplyDelivery (1) | | | | | | | | | | | | occurrence[x]
SupplyRequest (1) | identifier | | status | category | priority | orderedItem | | | | occurrence[x] | | authoredOn | requester | | supplier* | | | | | reason[x]
Task (2) | identifier* | | status | intent | priority | code focus | for | context | | | executionPeriod | authoredOn | agent | | performerType* owner | | | | | reason
Infrastructural Resources
AuditEvent (3) | | | | | | type subtype* action outcome outcomeDesc entity* identifier reference type | | role lifecycle securityLabel* name description query detail* type value | | | | recorded | | | | | source site identifier type* | agent* role* reference userId altId name requestor | location media network address type | purposeOfEvent* policy* purposeOfUse*
Basic (1) | identifier* | | | | | code | subject | | | | | created | author
Binary (5)
Bundle (5) | identifier | | | type
Composition (2) | identifier | | status | type class mode | | | subject | encounter | | | date | | author* | | | | party
Consent (1) | identifier | | status | category* | | | patient | | | | period | dateTime | | | consentingParty* actor* | | organization*
DocumentManifest (2) | masterIdentifier identifier* | | status | type | | | subject | | | | created | | author* | | | recipient*
DocumentReference (3) | masterIdentifier identifier* | | status docStatus | type class | | | subject | encounter | | | created | indexed | author* | | | | authenticator
Endpoint (2) | identifier* | | status | connectionType | | name | | | | | period
Linkage (0) | | | active | | | | | | | | | | author
List (1) | identifier* | | status | mode | | code | subject | encounter | | | | date | source
Media (1) | identifier* | | | type subtype view | | | subject | context | | | occurrence[x] | | | | operator | | | | | reasonCode*
MessageDefinition (0) | url identifier | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
MessageHeader (3) | | | | | | event | destination* receiver sender | | timestamp | | | | enterer author | | source | | responsible | | | reason
OperationOutcome (5)
Provenance (3) | | | | | | target* | | | | | period | recorded | | | who[x] | | | agent* | location | reason* activity
Questionnaire (3) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
QuestionnaireResponse (3) | identifier | | status | | | | subject | context | | | | authored | author | source
Subscription (3) | | | status | | | | contact* | | | | end | | | | | | | | reason
Conformance Resources
CapabilityStatement (3) | url | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
CodeSystem (5) | url identifier | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
CompartmentDefinition (1) | url | | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
ConceptMap (3) | url identifier | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
DataElement (1) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher
ExpansionProfile (2) | url identifier | version | status | experimental | | | | | | | | date | | | | | publisher
GraphDefinition (0) | url | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
ImplementationGuide (1) | url | version | status | experimental | | | | | | | | date | | | | | publisher
NamingSystem (1) | value | | status | kind | | | | | | | | date | | | | | publisher responsible
OperationDefinition (4) | url | version | status | kind experimental | | | | | | | | date | | | | | publisher | | | purpose
SearchParameter (3) | url | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
StructureDefinition (5) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
StructureMap (2) | url identifier* | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
TestReport (0) | identifier | | status | | | | | | | | issued
TestScript (2) | url identifier | version | status | experimental | | | | | | | | date | | | | | publisher | | | purpose
ValueSet (5) | url identifier* | version | status | experimental | | | | | lockedDate | | | date | | | | | publisher | | | purpose
Financial Resources
Account (2) | identifier* | | status | type | | name | subject | | | | period | active
ChargeItem (0) | identifier | | status | | | code | subject | context | | | occurrence[x] | | | | actor enterer | | | | | reason* service*
Claim (2) | identifier* | | status | type subType* use | | | patient | | | | billablePeriod serviced[x] | created | enterer | provider organization | provider | referral | | | facility location[x]
ClaimResponse (2) | identifier* | | status | | | | | | | | created | | requestProvider requestOrganization | | | | | request
Contract (1) | | | | | | | | | | | issued | | | | subject*
Coverage (2) | identifier* | | status | type | | | policyHolder subscriber subscriberId beneficiary | | | | period | | payor*
EligibilityRequest (2) | identifier* | | status | priority | | | patient | | | | serviced[x] | created | enterer | provider organization | | | | insurer | facility

EligibilityResponse (2) | identifier* | | status | | | | | | | | created | | requestProvider requestOrganization | | | | | request

EnrollmentRequest (0)

EnrollmentResponse (0)

ExplanationOfBenefit (2) | identifier* | | status | type subType* | | | patient | | | | billablePeriod serviced[x] | created | enterer insurer | provider organization | provider | referral | | | facility location[x] | claim

PaymentNotice (2) | identifier* | | status | | | | request response | | | | statusDate | created | | provider organization

PaymentReconciliation (2) | identifier* | | status | | | | payee | | | | period | created | | organization | | | | | request

Full

Resource | id | id.version | status | class | grade | what | who.focus | context | when.init | when.planned | when.done | when.recorded | who.author | who.source | who.actor | who.cause | who.witness | who | where | why

Clinical Resources

ActivityDefinition (2) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

AdverseEvent (0) | | | | category : code | | type : CodeableConcept | | | date : dateTime | | | | recorder : Patient|Practitioner|RelatedPerson | | | | | subject : Patient|ResearchSubject|Medication|Device

AllergyIntolerance (3) | identifier* : Identifier | | clinicalStatus : code verificationStatus : code | type : code category* : code | criticality : code | code : CodeableConcept | patient : Patient | | onset[x] : dateTime|Age|Period|Range|string | | | assertedDate : dateTime | recorder : Practitioner|Patient | asserter : Patient|RelatedPerson|Practitioner

BodySite (1) | identifier* : Identifier | | active : boolean | | | code : CodeableConcept qualifier* : CodeableConcept description : string image* : Attachment | patient : Patient

CarePlan (2) | identifier* : Identifier | | status : code | category* : CodeableConcept | | description : string | subject : Patient|Group | context : Encounter|EpisodeOfCare | period : Period | | | | | author* : Patient|Practitioner|RelatedPerson|Organization|CareTeam | careTeam* : CareTeam | | | | | addresses* : Condition

CareTeam (2) | identifier* : Identifier | | status : code | category* : CodeableConcept | | | subject : Patient|Group | context : Encounter|EpisodeOfCare | period : Period | | | | | | member : Practitioner|RelatedPerson|Patient|Organization|CareTeam | | | | | reasonCode* : CodeableConcept reasonReference* : Condition

ClinicalImpression (0) | identifier* : Identifier | | status : code | | | code : CodeableConcept | subject : Patient|Group | context : Encounter|EpisodeOfCare | | | effective[x] : dateTime|Period | date : dateTime | assessor : Practitioner | | | | | | | problem* : Condition|AllergyIntolerance

Condition (3) | identifier* : Identifier | | clinicalStatus : code verificationStatus : code | category* : CodeableConcept | severity : CodeableConcept | code : CodeableConcept | subject : Patient|Group | context : Encounter|EpisodeOfCare | onset[x] : dateTime|Age|Period|Range|string | | abatement[x] : dateTime|Age|boolean|Period|Range|string | assertedDate : dateTime | asserter : Practitioner|Patient|RelatedPerson | | | | | | code* : CodeableConcept detail* : Any

DetectedIssue (1) | identifier : Identifier | | status : code | category : CodeableConcept | severity : code | | patient : Patient | | | | | date : dateTime | author : Practitioner|Device

DiagnosticReport (3) | identifier* : Identifier | | status : code | category : CodeableConcept | | code : CodeableConcept | subject : Patient|Group|Device|Location | context : Encounter|EpisodeOfCare | | | effective[x] : dateTime|Period | issued : instant | | | | | performer* :

FamilyMemberHistory (2) | identifier* : Identifier | | status : code | | | | patient : Patient | | | | | date : dateTime | | | | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation|AllergyIntolerance|QuestionnaireResponse

Goal (2) | identifier* : Identifier | | status : code | category* : CodeableConcept | priority : CodeableConcept | description : CodeableConcept | subject : Patient|Group|Organization | | | start[x] : date|CodeableConcept | due[x] : date|Duration | statusDate : date | | expressedBy : Patient|Practitioner|RelatedPerson | | | | | | addresses* : Condition|Observation|MedicationStatement|NutritionOrder|ProcedureRequest|RiskAssessment

GuidanceResponse (2) | identifier : Identifier | | status : code | | | | subject : Patient|Group | context : Encounter|EpisodeOfCare | | | occurrenceDateTime : dateTime | | | | performer : Device | | | | | reason[x] : CodeableConcept|Reference(Any)

ImagingManifest (1) | identifier : Identifier | | | | | | patient : Patient | | | | | authoringTime : dateTime | author : Practitioner|Device|Organization|Patient|RelatedPerson

ImagingStudy (3) | uid : oid accession : Identifier identifier* : Identifier | | availability : code | modalityList* : Coding | | | patient : Patient | context : Encounter|EpisodeOfCare | started : dateTime | | | | | performer* : Practitioner | basedOn* : ReferralRequest|CarePlan|ProcedureRequest referrer : Practitioner | interpreter* : Practitioner | | | reason : CodeableConcept

Immunization (3) | identifier* : Identifier | | status : code notGiven : boolean | | | vaccineCode : CodeableConcept | patient : Patient | encounter : Encounter | date : dateTime | | | | | primarySource : boolean reportOrigin : CodeableConcept | actor : Practitioner | | | | location : Location

ImmunizationRecommendation (1) | identifier* : Identifier | | | | | | patient : Patient

Library (2) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | type : CodeableConcept | | | | | date : dateTime | | | | | | publisher : string | | | purpose : markdown

Measure (2) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | | purpose : markdown

MeasureReport (2) | identifier : Identifier | | status : code | type : code | | measure : Measure | patient : Patient | | | | | date : dateTime | | reportingOrganization : Organization

Medication (3) | | | | code : CodeableConcept isBrand : boolean isOverTheCounter : boolean | | | | | | | | | | | manufacturer : Organization

MedicationAdministration (2) | identifier* : Identifier | | status : code | | | medication[x] : CodeableConcept|Reference(Medication) | subject : Patient|Group | context : Encounter|EpisodeOfCare supportingInformation* : Any | | | effective[x] : dateTime|Period | | | | performer* : | | | | | reasonReference* : Condition|Observation

MedicationDispense (2) | identifier* : Identifier | | status : code | | | medication[x] : CodeableConcept|Reference(Medication) | subject : Patient|Group | supportingInformation* : Any

MedicationRequest (3) | identifier* : Identifier | | status : code | intent : code category : CodeableConcept | priority : code | medication[x] : CodeableConcept|Reference(Medication) | subject : Patient|Group | context : Encounter|EpisodeOfCare supportingInformation* : Any | | | | authoredOn : dateTime | requester : | | | | | recorder : Practitioner performer : Organization | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation

MedicationStatement (3) | identifier* : Identifier | | status : code | category : CodeableConcept | | medication[x] : CodeableConcept|Reference(Medication) | | | | | effective[x] : dateTime|Period | dateAsserted : dateTime | | informationSource : Patient|Practitioner|RelatedPerson|Organization | | | | subject : Patient|Group | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation

NutritionOrder (2) | identifier* : Identifier | | status : code | | | type* : CodeableConcept modifier : CodeableConcept type : CodeableConcept baseFormulaType : CodeableConcept additiveType : CodeableConcept | patient : Patient | encounter : Encounter | | | | dateTime : dateTime | orderer

: Practitioner

Observation (5) | identifier* : Identifier | | status : code | category* : CodeableConcept | | code : CodeableConcept code : CodeableConcept | subject : Patient|Group|Device|Location | context : Encounter|EpisodeOfCare | | | effective[x] : dateTime|Period | issued : instant | | | performer* : Practitioner|Organization|Patient|RelatedPerson

PlanDefinition (2) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

Procedure (3) | identifier* : Identifier | | status : code | category : CodeableConcept | | code : CodeableConcept | subject : Patient|Group | context : Encounter|EpisodeOfCare | | | performed[x] : dateTime|Period | | | | actor : Practitioner|Organization|Patient|RelatedPerson|Device | | | | location : Location | reasonCode* : CodeableConcept reasonReference* : Condition|Observation

ProcedureRequest (3) | identifier* : Identifier | | status : code | intent : code priority : code category* : CodeableConcept | | code : CodeableConcept | subject : Patient|Group|Location|Device | context : Encounter|EpisodeOfCare | | occurrence[x] : dateTime|Period|Timing | | authoredOn : dateTime | agent : Device|Practitioner|Organization | | performerType : CodeableConcept performer : Practitioner|Organization|Patient|Device|RelatedPerson|HealthcareService | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation

ReferralRequest (1) | identifier* : Identifier | | status : code | intent : code type : CodeableConcept | priority : code | serviceRequested* : CodeableConcept | subject : Patient|Group | context : Encounter|EpisodeOfCare | | occurrence[x] : dateTime|Period | | authoredOn : dateTime | agent : Practitioner|Organization|Patient|RelatedPerson|Device | | recipient* : Practitioner|Organization|HealthcareService | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation

RequestGroup (0) | identifier* : Identifier | | status : code | intent : code | priority : code | | subject : Patient|Group | context : Encounter|EpisodeOfCare | | | | authoredOn : dateTime | author : Device|Practitioner | | participant* : Patient|Person|Practitioner|RelatedPerson | | | | | reason[x] : CodeableConcept|Reference(Any)

ResearchStudy (0) | identifier* : Identifier | | status : code | | | focus* : CodeableConcept | enrollment* : Group | | | | period : Period | | | | sponsor : Organization principalInvestigator : Practitioner | | | | site* : Location | reasonStopped : CodeableConcept

ResearchSubject (0) | identifier : Identifier | | status : code | | | | | | | period : Period

RiskAssessment (1) | identifier : Identifier | | status : code | method : CodeableConcept | | code : CodeableConcept | subject : Patient|Group | context : Encounter|EpisodeOfCare | | | occurrence[x] : dateTime|Period | | | | performer : Practitioner|Device | | | | | reason[x] : CodeableConcept|Reference(Any)

Sequence (1) | identifier* : Identifier

ServiceDefinition (0) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

Specimen (2) | identifier* : Identifier accessionIdentifier : Identifier | | status : code | | | type : CodeableConcept | subject : Patient|Group|Device|Substance | | collected[x] : dateTime|Period | | receivedTime : dateTime | | | | collector : Practitioner | | | | | request* : ProcedureRequest

VisionPrescription (1) | identifier* : Identifier | | status : code | | | | patient : Patient | encounter : Encounter | | | | dateWritten : dateTime | prescriber : Practitioner | | | | | | | reason[x] : CodeableConcept|Reference(Condition)

Administrative Resources

Device (2) | identifier* : Identifier udi : | | status : code | | | deviceIdentifier : string name : string type : CodeableConcept lotNumber : string manufacturer : string manufactureDate : dateTime expirationDate : dateTime model : string version : string | patient : Patient | | | | | | | owner : Organization contact* : ContactPoint | | | | | location : Location url : uri

DeviceComponent (1) | identifier : Identifier | | | type : CodeableConcept | | | | | lastSystemChange : instant

DeviceMetric (1) | identifier : Identifier | | | type : CodeableConcept category : code

Group (1) | identifier* : Identifier | | active : boolean | type : code actual : boolean | | code : CodeableConcept

HealthcareService (2) | identifier* : Identifier | | active : boolean | category : CodeableConcept | | | | | | | | | | | | | | location* : Location

Location (3) | identifier* : Identifier | | status : code operationalStatus : Coding | mode : code type : CodeableConcept physicalType : CodeableConcept

Organization (3) | identifier* : Identifier | | active : boolean | type* : CodeableConcept

Patient (5) | identifier* : Identifier | | active : boolean

Person (2) | identifier* : Identifier | | active : boolean

Practitioner (3) | identifier* : Identifier | | active : boolean

PractitionerRole (2) | identifier* : Identifier | | active : boolean | | | | | | | | period : Period | | | | | | | | location* : Location

RelatedPerson (2) | identifier* : Identifier | | active : boolean | relationship : CodeableConcept | | | | | | period : Period

Substance (2) | identifier* : Identifier | | status : code | category* : CodeableConcept | | code : CodeableConcept

Workflow Resources

Appointment (3) | identifier* : Identifier | | status : code | specialty* : CodeableConcept priority : unsignedInt | | | | supportingInformation* : Any | start : instant | | end : instant | | | | | | | actor : Patient|Practitioner|RelatedPerson|Device|HealthcareService|Location

AppointmentResponse (3) | identifier* : Identifier | | | | | | | | start : instant | | end : instant | | | | | | actor : Patient|Practitioner|RelatedPerson|Device|HealthcareService|Location

Communication (2) | identifier* : Identifier | | status : code | category* : CodeableConcept | | | subject : Patient|Group recipient* : Device|Organization|Patient|Practitioner|RelatedPerson|Group | topic* : Any context : Encounter|EpisodeOfCare | sent : dateTime | | received : dateTime | | | | sender : Device|Organization|Patient|Practitioner|RelatedPerson | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation

CommunicationRequest (2) | identifier* : Identifier | | status : code | category* : CodeableConcept | priority : code | | subject : Patient|Group recipient* : Device|Organization|Patient|Practitioner|RelatedPerson|Group|CareTeam | topic* : Any context : Encounter|EpisodeOfCare | | occurrence[x] : dateTime|Period | | authoredOn : dateTime | agent : Practitioner|Organization|Patient|RelatedPerson|Device | | sender : Device|Organization|Patient|Practitioner|RelatedPerson | | | | | reasonCode* : CodeableConcept reasonReference* : Condition|Observation

DeviceRequest (0) | identifier* : Identifier | | status : code | intent : CodeableConcept | priority : code | code[x] : Device)|CodeableConcep | subject : Patient|Group|Location|Device | context : Encounter|EpisodeOfCare | | occurrence[x] : dateTime|Period|Timing | | authoredOn : dateTime | agent : Device|Practitioner|Organization | | performerType : CodeableConcept performer : Practitioner|Organization|Patient|Device|RelatedPerson|HealthcareService | | | | | reasonCode* : CodeableConcept reasonReference* : Any

DeviceUseStatement (0) | identifier* : Identifier | | status : code | | | | subject : Patient|Group | | | | whenUsed : Period timing[x] : Timing|Period|dateTime | recordedOn : dateTime | | | source : Patient|Practitioner|RelatedPerson device : Device | | | | | indication* : CodeableConcept

Encounter (2) | identifier* : Identifier | | status : code | class : Coding type* : CodeableConcept | priority : CodeableConcept | | subject : Patient|Group | episodeOfCare* : EpisodeOfCare | | | period : Period | | | | | | | individual : Practitioner|RelatedPerson | location : Location | reason* : CodeableConcept condition : Condition|Procedure

EpisodeOfCare (2) | identifier* : Identifier | | status : code | type* : CodeableConcept | | condition : Condition | patient : Patient | | period : Period

Flag (1) | identifier* : Identifier | | status : code | category : CodeableConcept | | code : CodeableConcept | subject : Patient|Location|Group|Organization|Practitioner|PlanDefinition|Medication|Procedure | period : Period encounter : Encounter | | | | | author : Device|Organization|Patient|Practitioner

ProcessRequest (2) | identifier* : Identifier | | status : code | action : code | | | target : Organization | | | | | created : dateTime | | provider : Practitioner organization : Organization

ProcessResponse (2) | identifier* : Identifier | | status : code | | | | | | | | created : dateTime | | organization : Organization | | | | | request : Any

Schedule (3) | identifier* : Identifier | | active : boolean | serviceCategory : CodeableConcept serviceType* : CodeableConcept specialty* : CodeableConcept | | | actor* : Patient|Practitioner|PractitionerRole|RelatedPerson|Device|HealthcareService|Location | | | planningHorizon : Period

Slot (3) | identifier* : Identifier | | | serviceCategory : CodeableConcept serviceType* : CodeableConcept specialty* : CodeableConcept appointmentType : CodeableConcept | | | | | start : instant end : instant

SupplyDelivery (1) | | | | | | | | | | | occurrence[x] : dateTime|Period|Timing

SupplyRequest (1) | identifier : Identifier | | status : code | category : CodeableConcept | priority : code | orderedItem : | | | | occurrence[x] : dateTime|Period|Timing | | authoredOn : dateTime | requester : | | supplier* : Organization | | | | | reason[x] : CodeableConcept|Reference(Any)

Task (2) | identifier* : Identifier | | status : code | intent : code | priority : code | code : CodeableConcept focus : Any | for : Any | context : Encounter|EpisodeOfCare | | | executionPeriod : Period | authoredOn : dateTime | agent : Device|Organization|Patient|Practitioner|RelatedPerson | | performerType* : CodeableConcept owner : Device|Organization|Patient|Practitioner|RelatedPerson | | | | | reason : CodeableConcept

Infrastructural Resources

AuditEvent (3) | | | | | | type : Coding subtype* : Coding action : code outcome : code outcomeDesc : string entity* : identifier : Identifier reference : Any type : Coding | | role : Coding lifecycle : Coding securityLabel* : Coding name : string description : string query : base64Binary detail* : type : string value : base64Binary | | | | recorded : instant | | | | | source : site : string identifier : Identifier type* : Coding | agent* : role* : CodeableConcept reference : Practitioner|Organization|Device|Patient|RelatedPerson userId : Identifier altId : string name : string requestor : boolean | location : Location media : Coding network : address : string type : code | purposeOfEvent* : CodeableConcept policy* : uri purposeOfUse* : CodeableConcept

Basic (1) | identifier* : Identifier | | | | | code : CodeableConcept | subject : Any | | | | | created : date | author : Practitioner|Patient|RelatedPerson

Binary (5)

Bundle (5) | identifier : Identifier | | | type : code

Composition (2) | identifier : Identifier | | status : code | type : CodeableConcept class : CodeableConcept mode : code | | | subject : Any | encounter : Encounter | | | date : dateTime | | author* : Practitioner|Device|Patient|RelatedPerson | | | | party : Patient|Practitioner|Organization

Consent (1) | identifier : Identifier | | status : code | category* : CodeableConcept | | | patient : Patient | | | | period : Period | dateTime : dateTime | | | consentingParty* : Organization|Patient|Practitioner|RelatedPerson actor* : | | organization* : Organization

DocumentManifest (2) | masterIdentifier : Identifier identifier* : Identifier | | status : code | type : CodeableConcept | | | subject : Patient|Practitioner|Group|Device | | | | created : dateTime | | author* : Practitioner|Organization|Device|Patient|RelatedPerson | | | recipient* : Patient|Practitioner|RelatedPerson|Organization

DocumentReference (3) | masterIdentifier : Identifier identifier* : Identifier | | status : code docStatus : code | type : CodeableConcept class : CodeableConcept | | | subject : Patient|Practitioner|Group|Device | encounter : Encounter | | | created : dateTime | indexed : instant | author* : Practitioner|Organization|Device|Patient|RelatedPerson | | | | authenticator : Practitioner|Organization

Endpoint (2) | identifier* : Identifier | | status : code | connectionType : Coding | | name : string | | | | | period : Period

Linkage (0) | | | active : boolean | | | | | | | | | | author : Practitioner|Organization

List (1) | identifier* : Identifier | | status : code | mode : code | | code : CodeableConcept | subject : Patient|Group|Device|Location | encounter : Encounter | | | | date : dateTime | source : Practitioner|Patient|Device

Media (1) | identifier* : Identifier | | | type : code subtype : CodeableConcept view : CodeableConcept | | | subject : Patient|Practitioner|Group|Device|Specimen | context : Encounter|EpisodeOfCare | | | occurrence[x] : dateTime|Period | | | | operator : Practitioner | | | | | reasonCode* : CodeableConcept

MessageDefinition (0) | url : uri identifier : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

MessageHeader (3) | | | | | | event : Coding | destination* : receiver : Practitioner|Organization sender : Practitioner|Organization | | timestamp : instant | | | | enterer : Practitioner author : Practitioner | | source : | | responsible : Practitioner|Organization | | | reason : CodeableConcept

OperationOutcome (5)

Provenance (3) | | | | | | target* : Any | | | | | period : Period | recorded : instant | | | who[x] : uri|Reference(Practitioner|RelatedPerson|Patient|Device|Organization) | | | agent* : | location : Location | reason* : Coding activity : Coding

Questionnaire (3) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

QuestionnaireResponse (3) | identifier : Identifier | | status : code | | | | subject : Any | context : Encounter|EpisodeOfCare | | | | authored : dateTime | author : Device|Practitioner|Patient|RelatedPerson | source : Patient|Practitioner|RelatedPerson

Subscription (3) | | | status : code | | | | contact* : ContactPoint | | | | end : instant | | | | | | | | reason : string

Conformance Resources

CapabilityStatement (3) | url : uri | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

CodeSystem (5) | url : uri identifier : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

CompartmentDefinition (1) | url : uri | | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

ConceptMap (3) | url : uri identifier : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

DataElement (1) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string

ExpansionProfile (2) | url : uri identifier : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string

GraphDefinition (0) | url : uri | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

ImplementationGuide (1) | url : uri | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string

NamingSystem (1) | value : string | | status : code | kind : code | | | | | | | | date : dateTime | | | | | publisher : string responsible : string

OperationDefinition (4) | url : uri | version : string | status : code | kind : code experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

SearchParameter (3) | url : uri | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

StructureDefinition (5) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

StructureMap (2) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

TestReport (0) | identifier : Identifier | | status : code | | | | | | | | | issued : dateTime

TestScript (2) | url : uri identifier : Identifier | version : string | status : code | experimental : boolean | | | | | | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

ValueSet (5) | url : uri identifier* : Identifier | version : string | status : code | experimental : boolean | | | | | lockedDate : date | | | date : dateTime | | | | | publisher : string | | | purpose : markdown

Financial Resources

Account (2) | identifier* : Identifier | | status : code | type : CodeableConcept | | name : string | subject : Patient|Device|Practitioner|Location|HealthcareService|Organization | | | | period : Period | active : Period

ChargeItem (0) | identifier : Identifier | | status : code | | | code : CodeableConcept | subject : Patient|Group | context : Encounter|EpisodeOfCare | | | occurrence[x] : dateTime|Period|Timing | | | | actor : Practitioner|Organization|Patient|Device|RelatedPerson enterer : Practitioner|Organization|Patient|Device|RelatedPerson | | | | | reason* : CodeableConcept service* : DiagnosticReport|ImagingStudy|Immunization|MedicationAdministration|MedicationDispense|Observation|Procedure|SupplyDelivery

Claim (2) | identifier* : Identifier | | status : code | type : CodeableConcept subType* : CodeableConcept use : code | | | patient : Patient | | | | billablePeriod : Period serviced[x] : date|Period | created : dateTime | enterer : Practitioner | provider : Practitioner organization : Organization | provider : Practitioner|Organization | referral : ReferralRequest | | | facility : Location location[x] : CodeableConcept|Address|Reference(Location)

ClaimResponse (2) | identifier* : Identifier | | status : code | | | | | | | | created : dateTime | | requestProvider : Practitioner requestOrganization : Organization | | | | | | request : Claim

Contract (1) | | | | | | | | | | | issued : dateTime | | | | subject* : Any

Coverage (2) | identifier* : Identifier | | status : code | type : CodeableConcept | | | policyHolder : Patient|RelatedPerson|Organization subscriber : Patient|RelatedPerson subscriberId : string beneficiary : Patient | | | | period : Period | | payor* : Organization|Patient|RelatedPerson

EligibilityRequest (2) | identifier* : Identifier | | status : code | priority : CodeableConcept | | | patient : Patient | | | | serviced[x] : date|Period | created : dateTime | enterer : Practitioner | provider : Practitioner organization : Organization | | | | insurer : Organization | facility : Location

EligibilityResponse (2) | identifier* : Identifier | | status : code | | | | | | | | created : dateTime | | requestProvider : Practitioner requestOrganization : Organization | | | | | | request : EligibilityRequest

EnrollmentRequest (0)

EnrollmentResponse (0)

ExplanationOfBenefit (2) | identifier* : Identifier | | status : code | type : CodeableConcept subType* : CodeableConcept | | | patient : Patient | | | | billablePeriod : Period serviced[x] : date|Period | created : dateTime | enterer : Practitioner insurer : Organization | provider : Practitioner organization : Organization | provider : Practitioner|Organization | referral : ReferralRequest | | | facility : Location location[x] : CodeableConcept|Address|Reference(Location) | claim : Claim

PaymentNotice (2) | identifier* : Identifier | | status : code | | | | request : Any response : Any | | | | statusDate : date | created : dateTime | | provider : Practitioner organization : Organization

PaymentReconciliation (2) | identifier* : Identifier | | status : code | | | | payee : Organization | | | | period : Period | created : dateTime | | organization : Organization | | | | | | request : ProcessRequest